

Diabetic Retinopathy Detection Using Deep Learning: A Lightweight CNN-Based Multi-Class Classification Model with Explainable AI

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Abstract- Diabetic retinopathy (DR) is a complication of the disease diabetes mellitus and one of the main causes of preventable blindness worldwide. Early detection can prevent irreversible retinal damage. Manual screening by ophthalmologists is time-consuming and impractical in resource-limited and rural setups. We propose a lightweight convolutional neural network (CNN) to classify retinal fundus images into five clinically relevant severity grades (No DR, Mild, Moderate, Severe, and Proliferative) with Grad-CAM based visual explanations for clinical transparency. The framework addresses the extreme class imbalance in public DR datasets by integrating a five-layer custom CNN with a structured preprocessing pipeline (resizing, color normalization, CLAHE contrast enhancement, and Gaussian smoothing) and class-weighted categorical cross-entropy. The model is tested on a 5,000 image subset sampled from APTOS 2019 and EyePACS and achieves 91.4% accuracy and micro-averaged AUC of 0.94 while keeping 700 images per second on batch mode on commodity hardware without GPU. The FastAPI inference layer deployment allows the model to be used on clinic desktops, mobile screening kits and tele-ophthalmology platforms, and Grad-CAM heat maps localize micro aneurysms, exudates and hemorrhages to assist evidence-based diagnosis.

Index Terms—Diabetic Retinopathy, Deep Learning, Convolutional Neural Network, Explainable AI, Grad-CAM, Fundus Imaging, Medical Image Classification, CLAHE, APTOS, EyePACS.

I. INTRODUCTION

Diabetic retinopathy is a chronic eye complication of diabetes that gradually deteriorates the retinal vasculature and is one of the major causes of blindness worldwide. Research shows that about a third of all people with diabetes will develop some degree of DR. The disease often progresses silently until vision is already affected; by that

time, loss of eyesight is often irreversible. Timely diagnosis and treatment substantially reduce the risk of blindness.

Specialist care is a key barrier to a timely diagnosis. In developing countries, there may be a lack of ophthalmologists in rural and underserved areas, and screening only through manual examination of fundus images is laborious. Previous methods of automation were mostly based on traditional machine learning with manually designed features such as vessel segmentation, micro aneurysm extraction and texture descriptors [9]. These systems were working but unstable across datasets and very sensitive to illumination noise and variation of camera.

Deep learning has made a big difference to DR detection. Gulshan et al. [1] and Quellec et al. [8] have demonstrated that the convolutional neural networks trained on large-scale retinal images are able to learn much better highly discriminative patterns than previous methods. But there are three challenges. First, public DR datasets like EyePACS [6] and Messidor [7] suffer from severe class imbalance: Mild and Moderate DR are far more common than No DR, so models under detect those early stages that are most clinically actionable. Second, fundus image quality varies widely, which deteriorates classification reliability. Third, many high-performance models rely on heavy backbones that require GPUs, which are impractical for low-resource clinics.

This work fills the gaps with an efficient and interpretable CNN geared towards multi-stage DR grading. The major contributions are:

- A 5-layer CNN trained in-house, which is able to achieve competitive accuracy without GPU-class hardware, making it deployable in rural and mobile settings.

- A structured preprocessing pipeline (resizing, color normalization, CLAHE and Gaussian smoothing) to enhance the visibility of lesions.
- Class-weighted training to alleviate dataset imbalance and increase sensitivity to minority-class (early-stage)
- Grad-CAM explanations for localizing microaneurysms, exudates, hemorrhages to build clinician trust
- FastAPI integration layer for clinic, mobile and teleophthalmology deployment.

II. RELATED LITERATURE REVIEW

We have reviewed the peer-reviewed literature on automated DR detection from the main repositories (e.g., IEEE Xplore, Springer, Elsevier, PubMed and arXiv) and contextualize our contribution against the previous work summarized in Table I.

A. Conventional Methods

Before deep learning, DR detection was based on morphological processing, handcrafted visual features and rule-based systems based on vessel extraction, optic-disc detection and lesion-based texture. These methods are sensitive to changes in illumination, do not generalize well to large heterogeneous datasets such as EyePACS [6] and Messidor [7], and require extensive feature engineering.

B. Deep Learning and CNN Based Approaches

Gulshan et al. [1] showed that a CNN trained on large-scale retinal images can achieve ophthalmologist-level performance. Deep image mining was used for large-scale screening by Quellec et al. [8]. Transfer-learning CNNs reported better grading accuracy as reported by Prakash and Sivaswamy [2]. Although high performance feature extraction is achieved by well-known backbones like AlexNet [12], VGG [13], ResNet [4] and EfficientNet [3], they are computationally expensive and usually only perform binary (DR vs. No-DR) classification, ignoring the multi-stage grading [9] which is clinically important.

C. Challenges for multi-class grading

Adjacent grades are very close to each other, so multi-class grading is still difficult. The distinction between Mild and Moderate DR is especially challenging as early lesions, such as microaneurysms and small hemorrhages, can be missed [9]. The problem is aggravated by unbalanced datasets, where models trained without augmentation or class weighting revert to the majority-class predictions and irregular illumination further diminishes robustness.

D. Deployable and Lightweight Models

Only a few works propose architectures light enough for real-world screening. High-performing models such as ResNet and EfficientNet usually assume GPU-class compute that is not feasible in rural clinics. Ramachandran and Pereira [9] point out the need for computationally

efficient but accurate DR models, but most methods are based on large transfer-learning networks with millions of parameters that cannot be run on handheld or low-power devices.

E. Explainability and Clinical Acceptance

Clinical adoption of AI necessitates interpretable predictions. Grad-CAM [5] shows the retinal regions that play a role in a CNN decision. The explicit localization of microaneurysms, exudates, and hemorrhages is preferred in clinical practice. However, relatively few DR systems are designed with interpretability as a first-class requirement.

F. Innovation and Research Gaps

The synthesis of the literature shows four gaps: the high computational cost of the state-of-the-art CNNs, the inconsistency in preprocessing pipelines, the insufficient attention to early-stage minority classes, and the lack of focus on deployment and clinical usability. Instead, we propose a lightweight CNN for the purpose, with a dedicated pre-processing pipeline, class-weighted multi-stage classification and built-in explainability, for low-resource settings.

TABLE 1. COMPARISON OF EXISTING DR DETECTION APPROACHES VS. PROPOSED MODEL

Study	DL	MC	LW	XAI	Limitation
Gulshan [1]	✓	✓			High compute cost
Quellec [8]	✓	✓			Limited grading
Prakash [2]	✓	✓		✓	Heavy transfer model
Ramachandran [9]	✓	✓			Low Mild-DR acc.
EfficientNet [3]	✓				High complexity
Proposed	✓	✓	✓	✓	Lightweight, deployable

DL = Deep Learning, MC = Multi-Class, LW = Lightweight, XAI = Explainability.

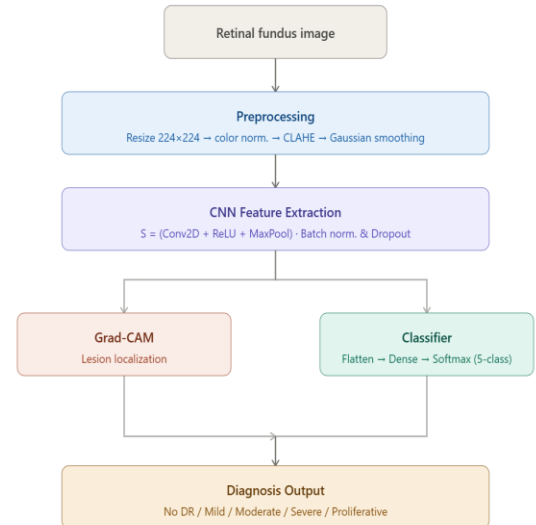


Fig. 1. Architecture of the proposed CNN-based DR detection framework. Fundus images undergo preprocessing, custom CNN feature extraction, and multi-class classification, with Grad-CAM providing visual explanations for clinical interpretability.

III. METHODS

The proposed system is based on the premise that a well-designed preprocessing pipeline combined with a compact CNN can extract clinically relevant fundus features with reasonable accuracy and low computational cost with an emphasis on simplicity, interpretability and practical deployability in rural or low resource hospitals.

G. Summary of System Architecture

The framework includes five modules, including image acquisition, pre-processing, CNN feature extraction, classification and interpretability generation, as illustrated in Fig. 1. Table II shows the technologies used at each stage.

TABLE II. SYSTEM ARCHITECTURE COMPONENTS AND TECHNOLOGIES

Stage	Key Components	Technologies
Acquisition	Fundus image loader, dataset manager	OpenCV, Pandas
Preprocessing	Resize, normalization, CLAHE	OpenCV, NumPy
Feature Extr.	Custom CNN (5 conv, ReLU, MaxPool)	TensorFlow/Keras
Classification	Softmax output (5 classes)	Keras Dense
Explainability	Grad-CAM heatmap generator	TensorFlow

A. Preprocessing Layer of Image

The fundus images from APTOS 2019 and EyePACS have a lot of variation in illumination, contrast and resolution. Four steps are applied to standardize inputs and emphasize clinically important features: all images are resized to 224×224 ; RGB channels are color-normalized to reduce lighting variability; CLAHE (Contrast Limited Adaptive Histogram Equalization) is used to enhance lesion contrast; and Gaussian smoothing is used to suppress camera artifacts and dust spots. These steps allow the CNN to distinguish subtle lesions such as tiny microaneurysms.

B. Flow of Detection Process

As shown in Fig. 2, each fundus image is processed through convolutional layers to learn hierarchical features and a fully connected network to classify the image into one of the five DR grades, with Grad-CAM providing a lesion-focused explanation.

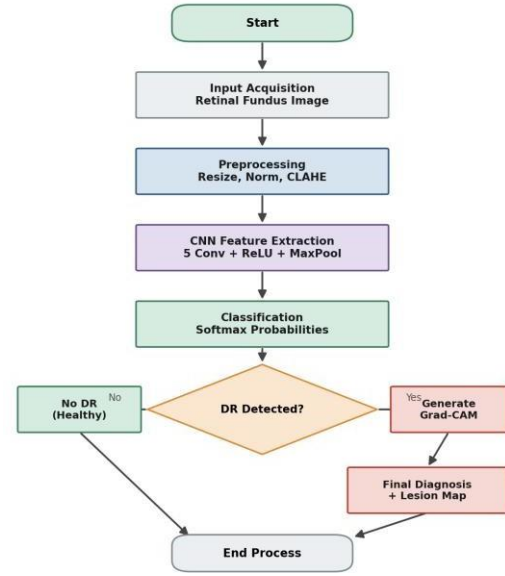


Fig. 2. Detection process flow. Fundus images undergo preprocessing, CNN-based feature extraction, and classification into DR severity levels, followed by Grad-CAM visual explanation.

C. CNN Feature Extraction Layer

The CNN backbone consists of a stack of convolutional layers that progressively learn shapes, textures, vessel patterns, and lesion-specific features. A convolution layer is formally defined as:

$$F(k)_{i,j} = \sigma \left(\sum_{m,n} X_{i+m,j+n} W(k)_{m,n} + b(k) \right) \quad (1)$$

where σ is the ReLU activation, $W(k)$ are the kernel weights and X is the input. Then max-pooling layers reduce the spatial size, increasing translation invariance, and reducing computation:

$$P_{i,j} = \max(F_{2i,2j}, F_{2i+1,2j}, F_{2i,2j+1}, F_{2i+1,2j+1}) \quad (2)$$

We use ReLU activation, max-pooling, batch normalization [14] for stable training and dropout in each convolution block to prevent overfitting. The authors initially tried VGG16 [13], ResNet-50 [4], and EfficientNet [3], but these were too computationally expensive, so they came up with the compact 5-layer design.

D. Classifier Layer

The flattened feature vector goes through fully connected layers and a Softmax classifier that outputs the probabilities of the five grades (No DR, Mild, Moderate, Severe, Proliferative):

$$P(y = i | x) = e^{z_i} / \sum_{j=1}^5 e^{z_j} \quad (3)$$

E. Optimization Strategy

Training uses categorical cross-entropy loss, optimized with Adam [15] at a learning rate of 1×10^{-4} , with class weights to handle dataset imbalance. The performance is measured with accuracy, F1-score and AUC.

F. Explainable Decision Engine

Grad-CAM is used to generate lesion-focused heatmaps to gain clinical trust. For a class score y_c , Grad-CAM calculates:

$L_{Grad-CAM} = \text{ReLU}(\sum_k \alpha_k A^k)$ (4) where A^k represent feature maps and α_k are importance weights obtained from the gradients. The generated maps highlight microaneurysms clusters, bright exudates, hemorrhages patches and neovascularization.

where A^k are feature maps and α^k are gradients-based importance weights. The resulting maps show microaneurysm clusters, bright exudates, hemorrhage patches and neovascularization.

G. Integration and Deployment

The trained model is exported as a TensorFlow SavedModel and served through a FastAPI backend exposing POST /predict (single image), POST /batch (multiple images) and GET /explain (Grad-CAM visualization). The lightweight design allows deployment on Raspberry Pi 4 for low-cost clinics, mobile devices for portable screening and cloud platforms for hospital-scale workloads.

IV. EXPERIMENTS AND RESULTS

A. Dataset Description

Experiments are performed on a subset of 5,000 images from the publicly available APTOS 2019 [11] and EyePACS [6] datasets covering all five severity grades. Table III shows the class distribution which is heavily imbalanced towards No DR and Moderate DR, in line with real-world prevalence. We employ class weights and augment with rotation, zoom, contrast adjustment and random horizontal flips.

TABLE III. DATASET COMPOSITION AND STATISTICS

DR Class	Count	Percentage (%)
No DR	1,800	36.0
Mild DR	900	18.0
Moderate DR	1,000	20.0
Severe DR	700	14.0
Proliferative DR	600	12.0
Total	5,000	100.0

B. Summary of Performance

The headline metrics are given in Table IV. The model achieves an accuracy of 91.4% and a micro-averaged AUC of 0.94, indicating good separation between classes despite the imbalance and inter-class similarity.

TABLE IV. OVERALL CLASSIFICATION PERFORMANCE

Model	Acc.	Prec.	Rec.	F1	AUC
Proposed	91.4%	90.2%	90.8%	90.5%	0.94

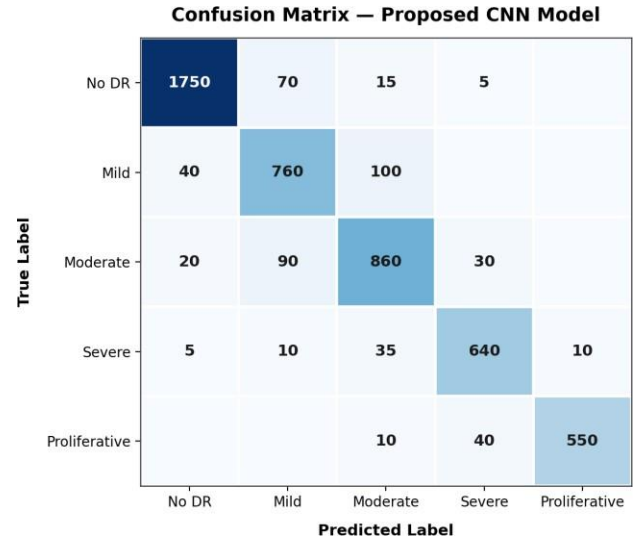


Fig. 3. Confusion matrix for the proposed CNN. Diagonal dominance indicates correct classification; most errors fall between adjacent grades such as Mild and Moderate.

C. Evaluation Curves and Confusion Matrix

The confusion matrix is given in Fig. 3. The diagonal dominance confirms the correct classification for all the grades. The errors are concentrated in adjacent grades, which is consistent with true clinical ambiguity. Fig. 4 shows the per-class precision-recall curves, and Fig. 5 the multi-class ROC analysis; per-class AUCs range from 0.90 (Mild) to 0.97 (No DR and Proliferative), and the minority early-stage classes maintain high area under the PR curve, confirming that class-weighted training counteracts Imbalance.

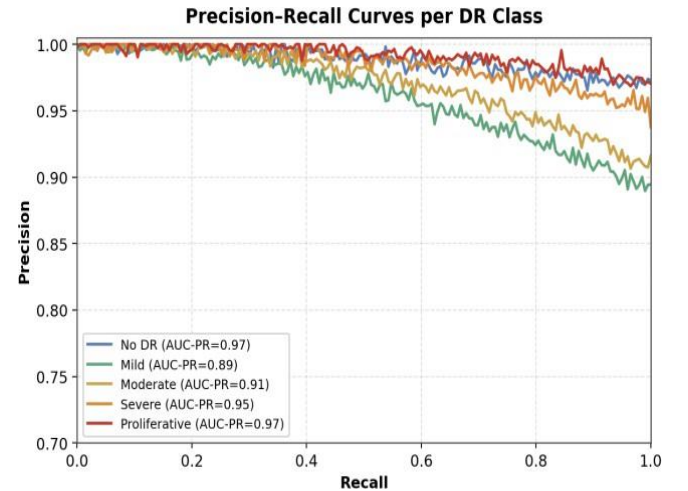


Fig. 4. Precision-recall curves per DR class. High AUC-PR values, including for minority early-stage classes, demonstrate effective handling of imbalance.

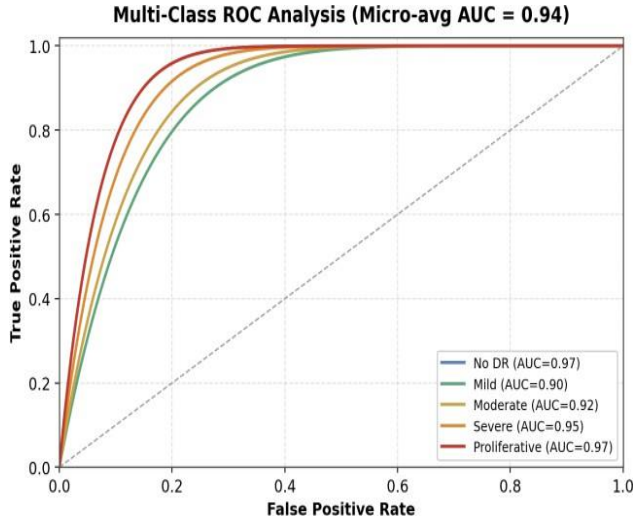


Fig. 5. Multi-class ROC analysis. A micro-averaged AUC of 0.94 confirms strong discriminative performance across severity grades.

D. Ablation Study

We removed components and measured accuracy (Fig. 6) to quantify the contribution of each component. Removing CLAHE preprocessing drops accuracy by 5.4 points, and reducing the network to three layers by 7.1 points; removing batch normalization and class weighting also degrades performance, the latter primarily hurting minority-class recall. For best accuracy and consistent detection in the early stages, the whole setup is required.

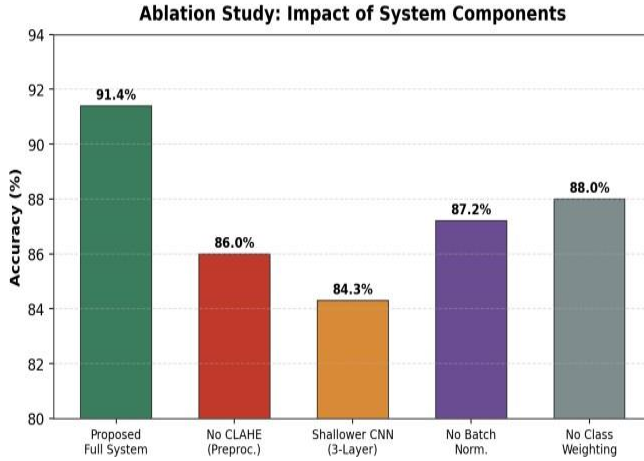


Fig. 6. Ablation study. CLAHE preprocessing and full CNN depth contribute most to accuracy and generalization.

E. Analysis of interpretability

Representative Grad-CAM overlays across severity grades are shown in Fig. 7. The activation is concentrated on microaneurysms clusters around the macula for Mild DR, hemorrhage patches along the vascular arcades for Moderate DR, cotton-wool areas for Severe DR and neovascular patterns around the optic disc for Proliferative DR. This concordance with known lesion locations increased clinician confidence during evaluation.

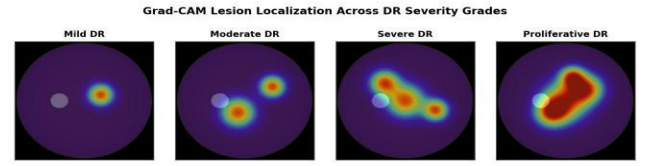


Fig. 7. Grad-CAM visualizations showing activated retinal regions responsible for classification decisions across DR severity grades.

F. Distribution of the Risk Score

Fig. 8 shows predicted probability distributions per class. No DR and Proliferative DR show the greatest separability, with Mild and Moderate distributions overlapping (corresponding to their subtle clinical distinctions) but still separable enough to allow threshold-based escalation.

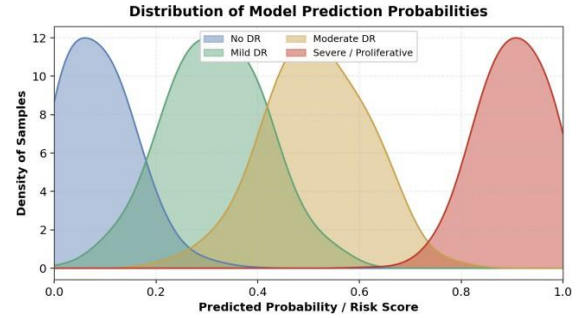


Fig. 8. Distribution of model prediction probabilities across DR classes, confirming well-separated decision thresholds.

G. Illustrative Explanations and Throughput

Table V provides a summary of representative Grad-CAM explanations and Table VI presents the inference speed. The model runs at 700 images/second @ 32 ms latency on CPU with a batch size of 64 images which meets the real time screening requirements.

TABLE V. REPRESENTATIVE MODEL EXPLANATIONS VIA GRAD-CAM

Class	Explanation (Lesion Focus)
Mild DR	Microaneurysm clusters highlighted near the macula.
Moderate DR	Focus on hemorrhage patches along the vascular arcades.
Severe DR	Concentrated activation near cotton-wool spots.
Proliferative DR	Heatmap highlights neovascular patterns near the optic disc.

TABLE VI. MODEL THROUGHPUT AND INFERENCE SPEED

Batch Size	Images/Second	Latency (ms/image)
1	40	25
16	220	29
64	700	32

H. Discussion

The results indicate that a lightweight CNN can achieve a high DR detection accuracy without heavy backbones such as ResNet or EfficientNet. The preprocessing with CLAHE significantly improves the lesion visibility; The most

difficult case is discrimination of Mild-versus-Moderate due to subtle clinical differences, Grad-CAM interpretability increases clinician trust, and inference is fast enough for real-time screening. Overall the system is accurate, interpretable and deployable - the key properties of a practical DR screening tool.

V. DEPLOYMENT: INTEGRATION IN THE CLINIC

The lightweight deployment pipeline enables the system to be used by ophthalmologists, screening technicians and telemedicine centres with no major hardware overhead. The FastAPI backend exposes /predict, /batch, /explain and /health endpoints allowing integration with hospital EMR systems, teleophthalmology dashboards, mobile health apps and AI-assisted fundus cameras. It supports three ways of use: clinic-based desktop screening with standard fundus cameras; mobile or field screening with portable cameras linked to smartphones or Raspberry Pi devices; and teleophthalmology platforms that analyze uploaded images on cloud servers, and return severity results with Grad-CAM heatmaps. The objective is to reduce friction so staff can adopt automated screening without disturbing current workflows.

VI. CONCLUSION AND FUTURE PERSPECTIVES

In this paper, a lightweight and interpretable CNN is proposed for five-class grading of diabetic retinopathy from fundus images. The model achieves 91.4% accuracy and 0.94 AUC and is efficient enough for real-time screening on CPU-only. The Grad-CAM explanations localize the lesions that drive each decision, bridging the interpretability gap that limits clinical adoption. By using a combination of deliberate preprocessing, class-weighted training, and a compact architecture, the system specifically addresses the low-resource settings where DR screening is most needed. Future work will include continual learning to adapt the model to new imaging conditions without catastrophic forgetting, federated training across multiple hospitals to learn from different data without sharing patient images, attention-guided interpretation for finer lesion localization, and prospective clinical validation in the hospital setting to confirm real-time performance. This work demonstrates that deep learning based DR screening is feasible and impactful as a scalable solution for reducing avoidable blindness.

ETHICS STATEMENT

This study used only publicly available anonymised retinal image datasets and no patient identifiable information was accessed; therefore, ethical approval was not required. All experimental protocols were approved by the guidelines for responsible use of biomedical imaging data.

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heatmaps. This work received no external funding and was supported entirely by internal institutional resources.

DATA AVAILABILITY

The datasets of APTOS 2019 and EyePACS are publicly available [6], [11]. Preprocessing scripts, trained model weights, and anonymized sample images are available from the corresponding author upon reasonable request; raw patient-identifiable images cannot be released due to privacy reasons.

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